HII Region Expansion: From Ionization Balance to the Modified Spitzer Solution

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Abstract

We present a self-contained derivation of the physics governing the expansion of HII regions around young OB stars, from ionization balance through the Strömgren sphere and the classic Spitzer thin-shell solution, to a modified formulation that correctly accounts for the shell mass physics. The analytic and numerical results are implemented in the open-source Python package `hii-expansion`, which integrates the thin-shell momentum equation in CGS units using `scipy` and `astropy`.

1 Introduction

Young massive OB stars emit copious ionizing photons ($h\nu > 13.6$ eV), carving out ionized nebulae—HII regions—in the surrounding interstellar medium. The rapid pressurization of the ionized gas drives a thin neutral shell outward, sweeping up ambient material in a process that shapes the structure of star-forming molecular clouds and regulates the efficiency of subsequent star formation. This note develops the physics of HII region expansion from first principles, beginning with ionization balance and the Strömgren sphere concept (Strömgren, 1939), proceeding through the dynamical thin-shell model of Spitzer (1978), and concluding with a modified formulation that corrects the treatment of the shell mass. All calculations are performed in CGS units using the Python package `hii-expansion` built on `scipy` (Virtanen et al., 2020) and `astropy` (Astropy Collaboration et al., 2022).

2 Ionization and Recombination Balance

In the interior of an HII region the gas is fully photoionized by the central star. In steady state the ionizing photon rate Q (photons s^{-1}) equals the total recombination rate within the ionized volume:

$$Q = 4\pi\alpha_B \int_0^R n(r)^2\, r^2\, dr, \tag{1}$$

where $n(r)$ is the hydrogen number density and α_B is the case B recombination coefficient, which accounts for recombinations to all levels except the ground state (Osterbrock & Ferland, 2006).

2.1 Case B recombination coefficient

We use the fitting formula of Draine (2011):

$$\alpha_B(T) = 2.54 \times 10^{-13}\, t_4^{-0.8163-0.0208\ln t_4} \quad \mathrm{cm^3\,s^{-1}}, \qquad t_4 \equiv \frac{T}{10^4\,\mathrm{K}}, \tag{2}$$

valid for $3\,000 \leq T \leq 30\,000$ K (Storey & Hummer, 1995). At $T = 10^4$ K, $\alpha_B \approx 2.54 \times 10^{-13}$ cm^3 s^{-1}.

2.2 Effective ionized density

Assuming that the ionization front propagates quasi-statically so that Equation (1) is satisfied instantaneously, the mean ionized density within radius R is

$$n_i(R) = \sqrt{\frac{3Q}{4\pi\alpha_B R^3}} \propto R^{-3/2}. \tag{3}$$

This quantity sets the interior gas pressure (Section 4).

3 The Strömgren Sphere

3.1 Uniform density

For a uniform ambient density n, Equation (1) gives an exact analytic result for the Strömgren radius:

$$R_{\mathrm{st}} = \left(\frac{3Q}{4\pi n^2 \alpha_B}\right)^{1/3}. \tag{4}$$

This scales as $R_{\mathrm{st}} \propto Q^{1/3} n^{-2/3} \alpha_B^{-1/3}$. For fiducial parameters $Q = 10^{49}$ s^{-1}, $n_0 = 100$ cm^{-3}, $T = 10^4$ K, Equation (4) yields $R_{\mathrm{st}} \approx 3.18$ pc.

3.2 Arbitrary density profile

For a radial density profile $n(r)$ the Strömgren radius is found numerically by root-finding on

$$f(R) = 4\pi\alpha_B \int_0^R n(r)^2 r^2\, dr - Q = 0 \tag{5}$$

using Brent's method, with the bracket expanded geometrically until $f(r_{\mathrm{hi}}) \geq 0$.

The effective rms density at the Strömgren radius is

$$n_{\mathrm{rms}} = \sqrt{\frac{3Q}{4\pi\alpha_B R_{\mathrm{st}}^3}}, \tag{6}$$

which equals n_0 for uniform density and is reduced for declining profiles. If $4\pi\alpha_B \int_0^\infty n^2 r^2\, dr < Q$ the medium is *density-bounded*: all ambient photons are used for ionization even at infinite radius and no finite Strömgren sphere exists.

3.3 Cored density profiles

Table 1 reports $R_{\rm st}$ and $n_{\rm rms}$ for the cored profile

$$n(r) = \frac{n_0}{1 + (r/r_0)^w} \tag{7}$$

with $r_0 \in \{1, 5, 10\}$ pc and $w \in \{1, 1.5, 2\}$. The $r_0 = 1$ pc, $w \geq 1.5$ cases are density-bounded.

Table 1: Strömgren radius and rms density for cored profiles. Parameters: $Q = 10^{49}$ s^{-1}, $n_0 = 100$ cm^{-3}, $T = 10^4$ K. "DB" = density-bounded.

r_0 [pc]	w	$R_{\rm st}$ [pc]	$n_{\rm rms}$ [cm^{-3}]	$n_{\rm rms}/n_0$
0 (uniform)	—	3.175	100.0	1.0000
1	1.0	15.308	9.44	0.0944
1	1.5		DB	
1	2.0		DB	
5	1.0	4.395	61.39	0.6139
5	1.5	4.081	68.60	0.6860
5	2.0	3.827	75.55	0.7555
10	1.0	3.729	78.55	0.7855
10	1.5	3.448	88.35	0.8835
10	2.0	3.310	93.94	0.9394

4 Thin-Shell Momentum Equation: Classic Spitzer Solution

4.1 Physical setup

Once the Strömgren sphere has formed, the ionized interior is overpressured relative to the neutral ambient medium and drives a thin shell outward. The interior pressure is (Spitzer, 1978)

$$P_{\rm in}(R) = 2 n_i(R)\, k_B T, \tag{8}$$

where the factor of 2 accounts for equal contributions from protons and electrons in a fully ionized hydrogen plasma, and $n_i(R)$ is given by Equation (3). The isothermal sound speed in the ionized gas is

$$c_{\rm II} = \sqrt{\frac{2 k_B T}{m_H}} \approx 12.8\ {\rm km\,s^{-1}} \quad (T = 10^4\ {\rm K}). \tag{9}$$

Note that $P_{\rm in} = n_i m_H c_{\rm II}^2$ is consistent with Equation (8) by construction.

4.2 ODE system

The momentum of the thin shell satisfies

$$\frac{d(M_{\rm sh}\dot{R})}{dt} = 4\pi R^2\Big(P_{\rm in} - n(R)\, m_H\, \dot{R}^2\Big)\,, \tag{10}$$

where the second term on the right is the ram-pressure deceleration from sweeping up ambient gas. Expanding the left-hand side and using the mass equation $\dot{M}_{\rm sh} = 4\pi R^2 n(R) m_H v$ (with $v \equiv \dot{R}$) yields the classic ODE system:

$$\dot{R} = v, \tag{11}$$

$$\dot{v} = \frac{4\pi R^2}{M_{\rm sh}}\big(P_{\rm in}(R) - n(R)\, m_H\, v^2\big)\,, \tag{12}$$

$$\dot{M}_{\rm sh} = 4\pi R^2\, n(R)\, m_H\, v. \tag{13}$$

The ODE state vector is $[R, v, M_{\rm sh}]$. Initial conditions at $t = 0$ are $R(0) = R_{\rm st}$, $v(0) = c_{\rm II}$ (D-critical front), and $M_{\rm sh}(0) = \frac{4\pi}{3} R_{\rm st}^3 n_0 m_H$ (all mass within $R_{\rm st}$ swept into the shell).

4.3 Quasi-static analytic solution

Neglecting the ram-pressure term and assuming instantaneous ionization balance, Equation (10) reduces to (Spitzer, 1978)

$$\dot{R} \approx c_{\rm II}\left(\frac{R_{\rm st}}{R}\right)^{3/4}, \tag{14}$$

with exact solution

$$R(t) = R_{\rm st}\left[1 + \frac{7}{4}\frac{c_{\rm II}}{R_{\rm st}}\, t\right]^{4/7}. \tag{15}$$

At late times the asymptotic power law is $R \propto t^{4/7}$.

The numerical ODE integrator (`scipy.integrate.solve_ivp`, RK45, rtol = 10^{-10}) reproduces the analytic result of Equation (15) to within $\sim 7\%$ at late times; the residual arises because the quasi-static solution neglects the ram-pressure term $n m_H v^2$.

4.4 Physical Origin and Magnitude of the Ram Pressure Term

Origin. As the shell sweeps outward at velocity v, it engulfs a column of ambient gas of surface density $n(R)\, m_H\, v\, dt$ per unit area in time dt. Bringing this mass from rest to the shell velocity v requires momentum $n(R)\, m_H\, v^2\, dt$ per unit area, which must be supplied by the net pressure force. Integrated over the shell surface $4\pi R^2$, the ram pressure force opposing the shell motion is

$$F_{\rm ram} = 4\pi R^2\, n(R)\, m_H\, v^2. \tag{16}$$

This term appears explicitly in Equation (10) and in the ODE system (11).

Ram pressure is never negligible: the quasi-static identity. The ratio of the ram pressure force to the thermal driving force $F_{\rm th} = 4\pi R^2 P_{\rm in}$ is

$$\frac{F_{\rm ram}}{F_{\rm th}} = \frac{n(R)\, m_H\, v^2}{2\, n_i(R)\, k_B T}. \tag{17}$$

Along the Spitzer quasi-static trajectory $v = c_{\rm II}(R_{\rm st}/R)^{3/4}$, with $n_i \propto R^{-3/2}$ for uniform density, both numerator and denominator scale identically as $R^{-3/2}$, giving

$$\left.\frac{F_{\rm ram}}{F_{\rm th}}\right|_{\rm Spitzer} = \frac{n_0\, m_H\, c_{\rm II}^2}{2\, n_0\, k_B T} = \frac{m_H c_{\rm II}^2}{2\, k_B T} = 1 \quad \text{(at all times)}. \tag{18}$$

This result is not a coincidence: the quasi-static approximation is precisely the condition $P_{\rm in} = n\, m_H\, v^2$ (zero net force, $\dot{v} = 0$), which is identical to $F_{\rm ram}/F_{\rm th} = 1$. *Ram pressure is therefore never negligible; it is always exactly equal to the thermal driving force along the Spitzer trajectory.*

Why the no-ram ODE diverges from Spitzer. Disabling the ram pressure term in the ODE does *not* recover the Spitzer quasi-static solution (Figure 1). In fact, the no-ram ODE diverges from Spitzer regardless of the initial velocity v_0:

- **D-critical initial condition** ($v_0 = c_{\rm II}$): the full ODE has zero net force at $t = 0$ (quasi-static balance), but the no-ram ODE sees an immediate unbalanced force $4\pi R_{\rm st}^2 P_{\rm in}(R_{\rm st})$, which inflates the trajectory permanently.
- **Static initial condition** ($v_0 = 0$): the force at $t = 0$ is identical (since $F_{\rm ram} = 0$ when $v = 0$), and the initial acceleration $\dot{v}(0) = 3c_{\rm II}^2/R_{\rm st} = 3c_{\rm II}/T_{\rm dyn}$ brings the shell to $v \gg c_{\rm II}$ within $\sim T_{\rm dyn}$. Without a restoring deceleration, the shell coasts at this large velocity.

In both cases the shell asymptotically coasts at constant velocity ($R \propto t$, log-log slope $\to 1$), because $P_{\rm in} \propto R^{-3/2}$ delivers a *finite* total impulse $\Delta v = \int (4\pi R^2 P_{\rm in}/M_{\rm sh})\, dt < \infty$ while no deceleration mechanism exists.

Origin of the $\sim 7\%$ offset. The $\sim 7\%$ late-time offset between the full ODE (with ram pressure, $v_0 = c_{\rm II}$) and Spitzer arises not from ram pressure being large or small, but from the *inertial term* $M_{\rm sh}\, \dot{v}$. The Spitzer quasi-static derivation sets $\dot{v} = 0$ throughout; the full ODE retains a small but non-zero $\dot{v}$ that accumulates over time to produce the offset.

4.5 Power-law density profiles

For $n(r) = n_0(r_0/r)^w$ the asymptotic expansion law is (Franco et al., 1990)

$$R \propto t^{4/(7-2w)}, \qquad w < \tfrac{3}{2}. \tag{19}$$

The expansion accelerates with steeper profiles because the swept-up mass grows more slowly.

5 Modified Spitzer Solution

5.1 Motivation and corrections

The classic ODE system (Section 4) contains two physically inconsistent assumptions:

1. **Non-zero initial shell mass.** The classic formulation sets $M_{\rm sh}(0) = \frac{4\pi}{3} R_{\rm st}^3 n_0 m_H$, counting gas already ionized within $R_{\rm st}$ as part of the neutral shell. This gas is in the hot ionized interior and does not contribute to the shell momentum. The correct initial shell mass is $M_{\rm sh}(0) = 0$.

2. **Ionization mass exchange.** As the ionization front advances, neutral gas swept into the shell is continuously re-ionized at the front and transferred to the bubble interior, reducing the shell mass at every instant.

5.2 Derivation of the modified mass equation

At time t (shell radius R) the neutral shell mass equals the total mass swept from $R_{\rm st}$ to R minus the additional ionized mass $\Delta M_{\rm ion}$ that has been transferred to the interior:

$$M_{\rm sh}(R) = 4\pi m_H \int_{R_{\rm st}}^{R} n(r) r^2 \, dr - \Delta M_{\rm ion}. \tag{20}$$

The additional ionized mass is the difference between the current ionized bubble mass and the initial Strömgren sphere mass:

$$\Delta M_{\rm ion} = \frac{4\pi}{3} m_H \left[R^3 n_i(R) - R_{\rm st}^3 n_i(R_{\rm st}) \right]. \tag{21}$$

Differentiating $\Delta M_{\rm ion}$ with respect to time, and using $n_i \propto R^{-3/2}$ so that $d(R^3 n_i)/dR = R^2 n_i \cdot (3 - 3/2) = (3/2) R^2 n_i$, gives

$$\Delta \dot{M}_{\rm ion} = 4\pi m_H \frac{d}{dt}\left[\frac{R^3 n_i(R)}{3}\right] = \frac{4\pi}{3} m_H \cdot \frac{3}{2} R^2 n_i(R)\, v = 2\pi R^2 m_H n_i(R)\, v. \tag{22}$$

The modified shell-mass equation is therefore

$$\dot{M}_{\rm sh} = 4\pi R^2 n(R) m_H v - 2\pi R^2 m_H n_i(R) v = 2\pi R^2 m_H v [2n(R) - n_i(R)]. \tag{23}$$

The momentum equation (Equation 10) is unchanged: ionized gas that leaves the shell does so at the shell velocity, so the momentum-flux terms cancel exactly.

5.3 Modified ODE system

The modified ODE state vector $[R, v, M_{\rm sh}]$ satisfies

$$\dot{R} = v, \tag{24}$$

$$\dot{v} = \frac{4\pi R^2}{M_{\rm sh}} \left(P_{\rm in}(R) - n(R)\, m_H\, v^2 \right), \tag{25}$$

$$\dot{M}_{\rm sh} = 2\pi R^2 m_H v [2n(R) - n_i(R)]. \tag{26}$$

Initial conditions: $R(0) = R_{\rm st}$, $v(0) = c_{\rm II}$, $M_{\rm sh}(0) \approx 0$.

At $t = 0$ the net driving force is

$$F_0 = 4\pi R_{\rm st}^2 \big(P_{\rm in}(R_{\rm st}) - n(R_{\rm st})\, m_H\, c_{\rm II}^2\big) = 4\pi R_{\rm st}^2\, m_H\, c_{\rm II}^2 \left[n_i(R_{\rm st}) - n(R_{\rm st})\right]. \tag{27}$$

For a uniform medium $n_i(R_{\rm st}) = n_0$ exactly, so $F_0 = 0$ and neither shell accelerates initially. For a declining profile ($w > 0$) we showed in Section 2 that $n_i(R_{\rm st}) = n(R_{\rm st})\sqrt{3/(3-2w)} > n(R_{\rm st})$, giving $F_0 > 0$. The numerical integrator uses a seed mass $\sim 10^{-10}$ times a characteristic swept mass to avoid a literal divide-by-zero without affecting the dynamics.

5.4 Early-time overshoot and late-time crossover for declining profiles

For a declining density profile ($w > 0$) the modified and classic solutions exhibit a characteristic *crossover*: the modified solution expands faster than the classic at early times but slower at late times. The mechanism has two parts.

Early-time advantage. At $t = 0$ both shells start at $R_{\rm st}$ with $v_0 = c_{\rm II}$, but their inertias differ enormously: $M_{\rm sh}^{\rm mod} \approx 0$ versus $M_{\rm sh}^{\rm class} = 4\pi m_H \int_0^{R_{\rm st}} n(r) r^2\, dr \gg 0$. The positive overpressure force $F_0 > 0$ therefore accelerates the modified shell to $v \gg c_{\rm II}$ almost instantaneously, while the classic shell, burdened by its large initial mass, barely accelerates. The modified shell thus overshoots the quasi-static Spitzer trajectory $R_{\rm qs}(t) \propto t^{4/(7-2w)}$ at early times.

Late-time momentum deficit. Taking the time derivative of $M_{\rm sh}\, v$ using each ODE system:

$$\left.\frac{d(M_{\rm sh} v)}{dt}\right|_{\rm class} = 4\pi R^2 P_{\rm in}, \tag{28}$$

$$\left.\frac{d(M_{\rm sh} v)}{dt}\right|_{\rm mod} = 4\pi R^2 P_{\rm in} - 2\pi R^2 n_i m_H v^2. \tag{29}$$

The extra term $-2\pi R^2 n_i m_H v^2$ represents momentum carried away from the shell by ionized gas that is continuously transferred to the bubble interior at velocity v. This momentum sink accumulates over time, leaving the modified shell with less total momentum than the classic shell. Once the driving pressure $P_{\rm in} \propto R^{-3/2}$ weakens at large R, the modified shell's reduced momentum causes it to decelerate faster and fall behind the classic trajectory.

For $w = 1.0$ specifically, the ratio $n_i(R_{\rm st})/n(R_{\rm st}) = \sqrt{3} \approx 1.73$ gives a significant initial overpressure, making the crossover clearly visible (Figure 3). For $w = 0$ (uniform) the ratio equals unity, $F_0 = 0$, and no crossover occurs.

5.5 Late-time asymptotic behavior

At late times $R \gg R_{\rm st}$ both $n_i \ll n$ and $M_{\rm sh}^{\rm mod} < M_{\rm sh}^{\rm classic}$, but both systems converge to the same asymptotic power-law slope $R \propto t^{4/(7-2w)}$ because in this limit $n_i \to 0$ and the modified mass equation approaches the classic one, $\dot{M}_{\rm sh} \to 4\pi R^2 n m_H v$. The two solutions differ only in the pre-factor set during the transient phase.

6 Example Results

All numerical results use fiducial parameters $Q = 10^{49}$ s^{-1}, $n_0 = 100$ cm^{-3}, $T = 10^4$ K, giving $R_{\rm st} \approx 3.18$ pc, $c_{\rm II} \approx 12.8$ km s^{-1}, and $T_{\rm dyn} \equiv R_{\rm st}/c_{\rm II} \approx 0.24$ Myr.

6.1 Uniform density

Figure 2 shows the numerical thin-shell ODE solution alongside the Spitzer analytic formula (Equation 15) for uniform density. The ODE solution (solid) and analytic result (dashed) agree well; the $\sim 7\%$ offset at late times reflects the ram-pressure term neglected in the quasi-static derivation. Both curves asymptote to the reference $R \propto t^{4/7}$ slope (dotted).

6.2 Power-law density profiles

Figure 3 shows the expansion for $n(r) = n_0(r_0/r)^w$ with $w \in \{0, 0.5, 1.0, 1.4\}$. Steeper profiles lead to faster late-time expansion, consistent with the $R \propto t^{4/(7-2w)}$ scaling of Franco et al. (1990). For $w = 1.0$ the modified solution (dashed) lies *above* the classic solution at early times but *below* it at late times. This crossover is a direct consequence of the initial zero-mass overshoot and late-time momentum deficit described in Section 5.4. For $w \leq 0.5$ the initial overpressure ($n_i/n = \sqrt{3/(3-2w)}$) is small enough that the crossover, if present, is not visible on these axes. For $w = 1.4$ the modified ODE is ill-conditioned at $R_{\rm st}$ because $n_i \approx 3.87\,n > 2n$, so $\dot{M}_{\rm sh}(0) < 0$; no stable shell forms initially and the modified solution is not shown.

6.3 Cored density profiles

Figure 4 shows results for $n(r) = n_0/[1 + (r/r_0)^w]$ with $r_0 \in \{1, 5, 10\}$ pc and $w \in \{1, 1.5, 2\}$. Cases with $r_0 = 1$ pc and $w \geq 1.5$ are density-bounded (Table 1) and are excluded.

6.4 Classic vs. modified Spitzer solution

Figure 5 compares the classic and modified ODE systems for uniform density over $50\,T_{\rm dyn}$. The top panel shows $R(t)$: the two solutions are nearly indistinguishable, confirming that the shell-mass correction has negligible effect on the radius evolution. The middle panel shows $v(t)$: again nearly identical. The bottom panel shows $M_{\rm sh}(t)$: the modified ODE gives a systematically lower shell mass at all times, as expected from the zero initial shell mass and the continuous mass transfer to the ionized interior.

7 Conclusions

We have derived the equations governing HII region expansion from first principles and implemented them in the `hii-expansion` Python package. The key results are:

- The Strömgren radius is given analytically for uniform density (Equation 4) and numerically for arbitrary profiles (Equation 5).

- The classic Spitzer thin-shell ODE reproduces the $R \propto t^{4/7}$ analytic solution (Equation 15) within $\sim 7\%$ at late times.
- The modified ODE (Equations 24–23) correctly initializes with zero shell mass and accounts for continuous mass exchange at the ionization front, yielding a lower shell mass while leaving $R(t)$ and $v(t)$ nearly unchanged.
- Power-law profiles $n \propto r^{-w}$ accelerate the late-time expansion to $R \propto t^{4/(7-2w)}$ for $w < 3/2$.

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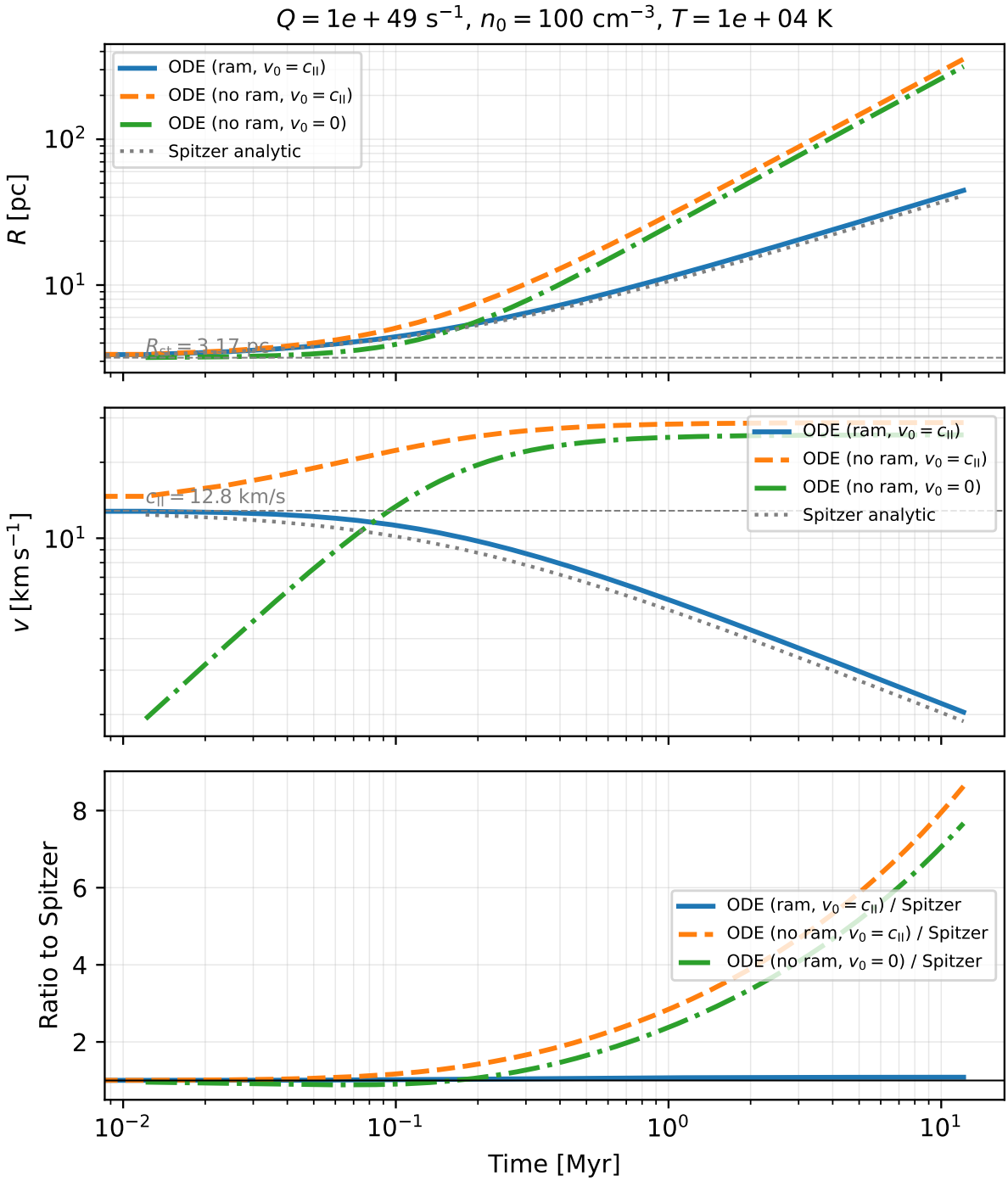

Figure 1: Effect of the ram pressure term and initial condition for uniform density ($n_0 = 100$ cm^{-3}). *Solid blue*: full ODE with ram pressure, D-critical IC ($v_0 = c_{\rm II}$); *dashed orange*: no-ram ODE, D-critical IC; *dash-dot green*: no-ram ODE, static IC ($v_0 = 0$); *dotted gray*: Spitzer quasi-static analytic. Both no-ram ODEs coast ($R \propto t$) and diverge from Spitzer because ram pressure equals $P_{\rm in}$ along the quasi-static trajectory and can never be neglected. The $\sim 7\%$ offset of the full ODE from Spitzer originates from the retained inertial term $M_{\rm sh}\dot{v}$.

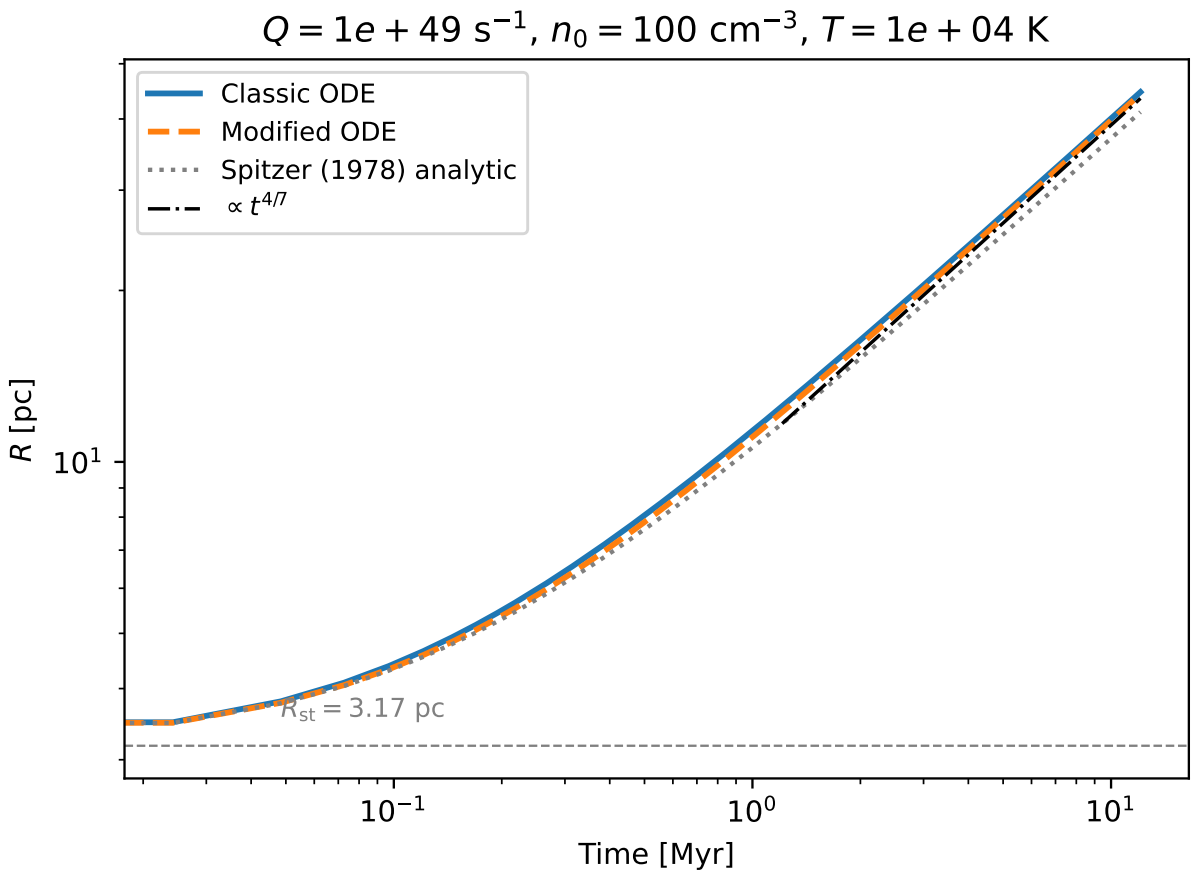

Figure 2: HII region expansion in a uniform medium ($n_0 = 100$ cm^{-3}). Solid: classic thin-shell ODE; dashed: Spitzer (1978) analytic solution; dotted: $R \propto t^{4/7}$ reference.

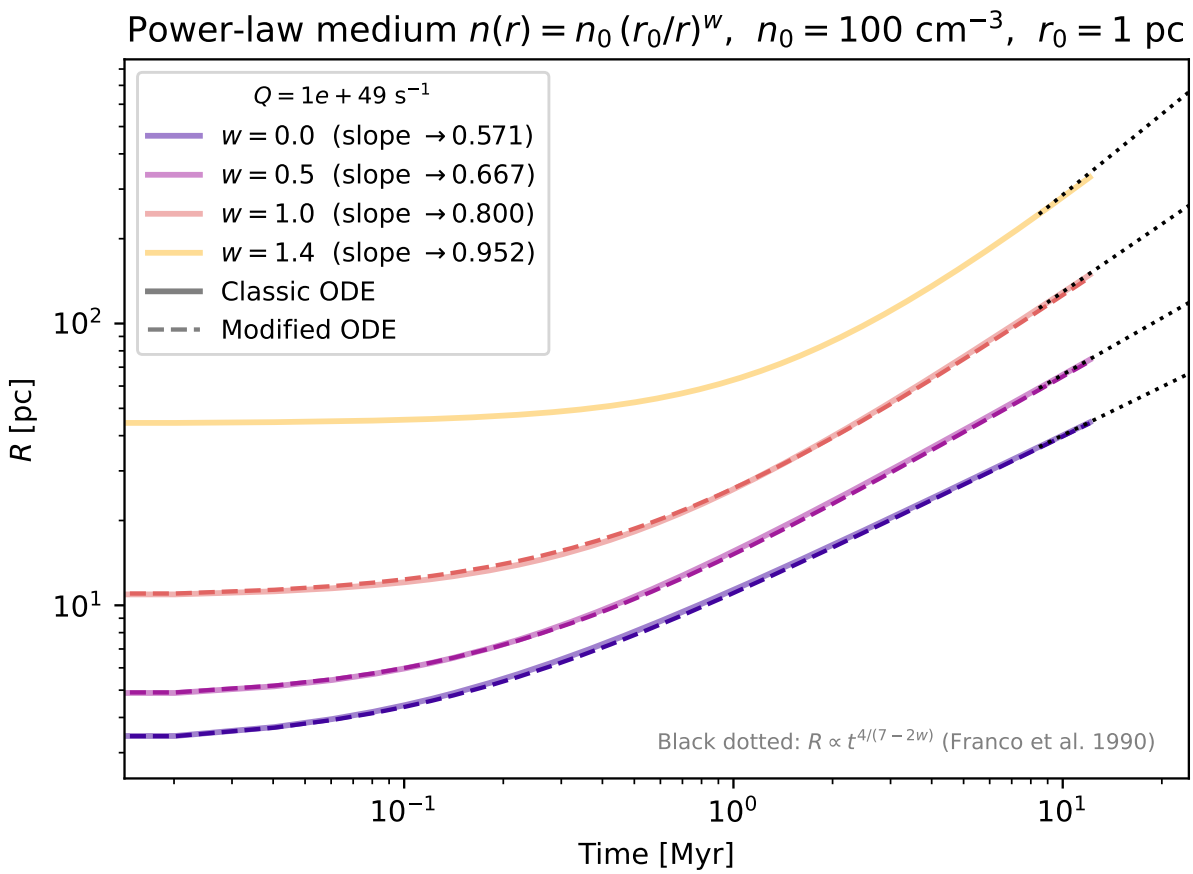

Figure 3: HII region expansion for power-law density profiles $n(r) = n_0(r_0/r)^w$. Reference slopes $t^{4/(7-2w)}$ from Franco et al. (1990) are overlaid.

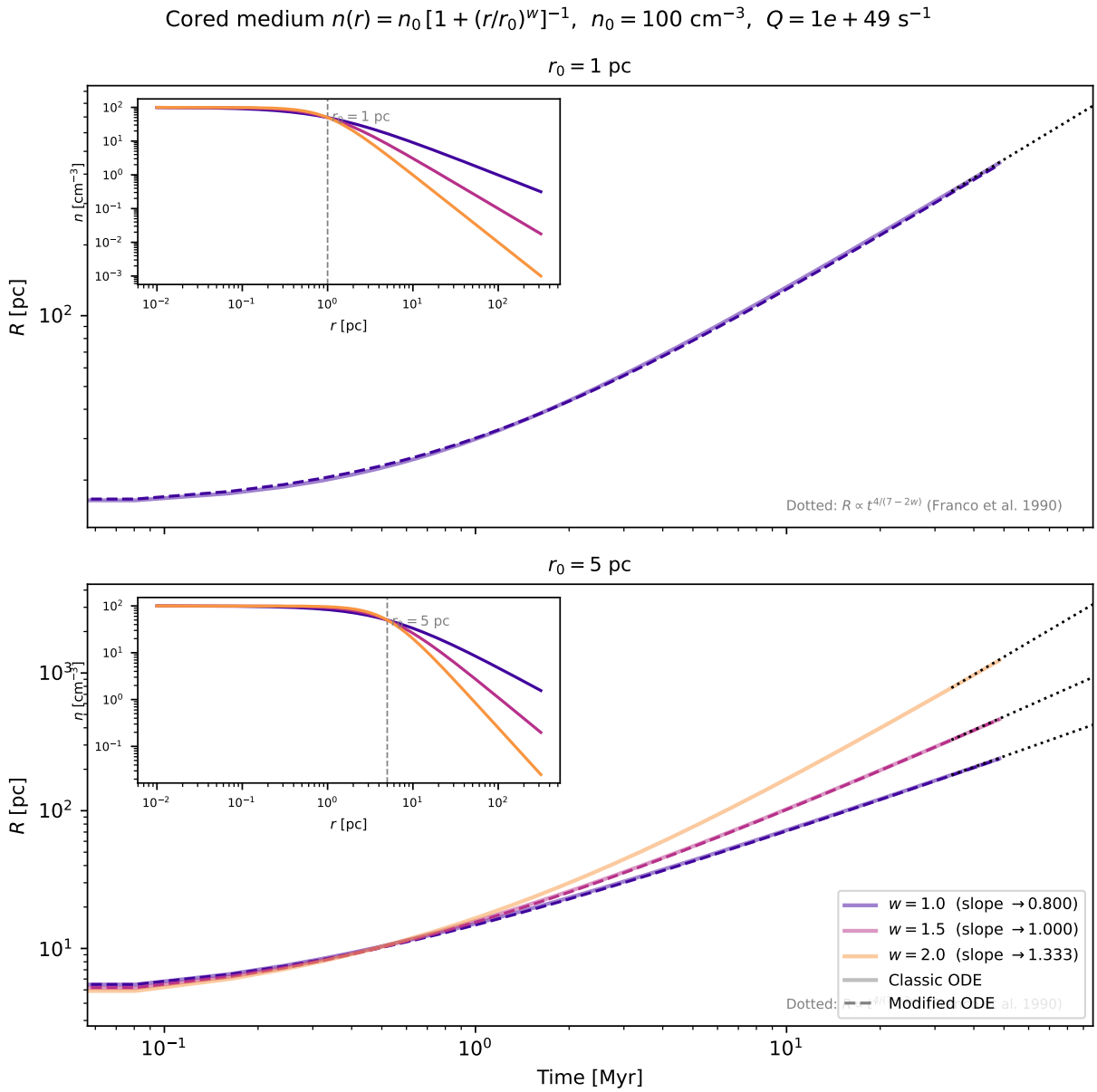

Figure 4: HII region expansion for cored profiles $n(r) = n_0/[1 + (r/r_0)^w]$. Density-bounded cases are annotated.

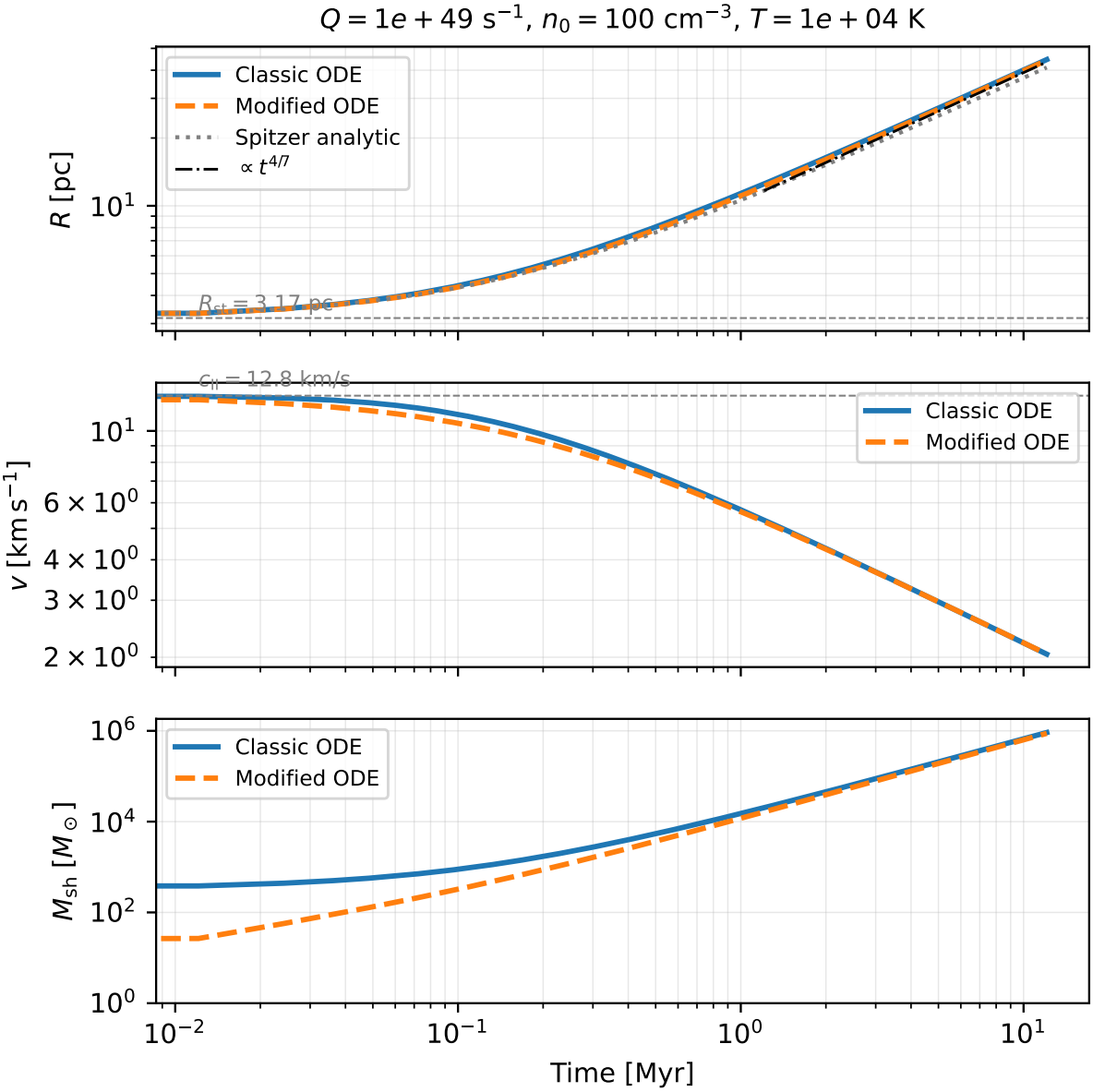

Figure 5: Comparison of the classic (solid, blue) and modified (dashed, orange) thin-shell ODE solutions for uniform density. Top to bottom: shell radius $R(t)$, shell velocity $v(t)$, shell mass $M_{\rm sh}(t)$. The radius and velocity agree closely; the shell mass in the modified solution is lower throughout.